Convergence and Taylor Series — Inconclusive Tests and Omitted Endpoints

Answer Key

AP Calculus BC

Quick Answers

1. 4	5. 1, 1
2. 0	6. -1, 5
3. 3	7. $-\log(2)$
4. 1	8. —

Curriculum

Level: AP Calculus BC
LIM-7, LIM-8 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

3. *If they got 6: divided the first term by the ratio instead of by one minus the ratio*
4. *If they got 0.5: reported the first term as the sum of the whole series*
6. *If they got 3: gave the radius itself as an endpoint, forgetting to shift by the centre*
7. *If they got 0.693: kept the alternating sum positive - the first term of this series is negative, so the sum is negative*

1. For $a_n = \dfrac{4n+1}{n+7}$, find $\lim a_n$ and apply the nth-term test.

Step 1: choose the tool. The nth-term test asks only what the terms do, so compute the limit of a_n before considering any other test — if it is not zero, no further work is needed.

Step 2: divide numerator and denominator by the highest power of n: $\dfrac{4+1/n}{1+7/n} \to \dfrac{4+0}{1+0}$.

Step 3: the limit is

$$\boxed{4}$$

Step 4: state the conclusion carefully. Since $\lim a_n = 4 \neq 0$, the series $\sum a_n$ **diverges** by the nth-term test. This is the one direction the test can prove.

2. Devi says $\lim 1/\sqrt{n} = 0$ implies $\sum 1/\sqrt{n}$ converges. Fix the reasoning.
Step 1: check Devi's arithmetic first — it is not the problem. As $n \to \infty$, $\sqrt{n} \to \infty$, so

$$\boxed{0}$$

Step 2: state what the test licenses. The nth-term test is a one-way implication: if $\lim a_n \neq 0$ the series diverges. It says *nothing* when the limit is 0. A limit of 0 is necessary for convergence and nowhere near sufficient — so Devi's step is not a weak argument, it is a test used in a direction it does not have.
Step 3: settle the series with a test that can. Written as $1/n^p$, this is a p-series with p equal to one half, and a p-series converges only for $p > 1$.
Step 4: verdict — $\sum 1/\sqrt{n}$ **diverges**, even though its terms go to zero. This example is worth remembering precisely because it is the counterexample to Devi's rule.

3. The ratio test proves $\sum_{n=0}^{\infty} 2(1/3)^n$ converges; the student's sum of 6 is wrong. Find the exact sum.
Step 1: name the misremembered formula. The student computed first term divided by ratio, $2 \div (1/3) = 6$. The geometric sum is first term divided by *one minus* the ratio.
Step 2: read off the parts. Because the index starts at $n = 0$, the first term is $a = 2$ and the ratio is $r = 1/3$, with $|r| < 1$.
Step 3: apply $\dfrac{a}{1-r} = \dfrac{2}{1-1/3} = \dfrac{2}{2/3}$:

$$\boxed{3}$$

Step 4: sanity-check against the partial sums: 2, then 2.67, then 2.89 — climbing toward 3 from below and never past it. A proposed sum of 6 is larger than twice the first term of a series with positive decreasing terms, which the partial sums make visibly impossible.

4. The ratio test returns 1 for $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$. Find the exact sum by telescoping.
Step 1: recognise why another tool is needed. A ratio-test limit of exactly 1 is the inconclusive case: it rules nothing in and nothing out, so the series must be handled by a method that produces the partial sums themselves.
Step 2: split by partial fractions: $\dfrac{1}{n(n+1)} = \dfrac{1}{n} - \dfrac{1}{n+1}$.
Step 3: write the partial sum and cancel. Every interior term appears twice with opposite signs, leaving only the ends: $S_N = 1 - \dfrac{1}{N+1}$. Let $N \to \infty$:

$$\boxed{1}$$

Step 4: locate the classmate's error. The value 0.5 is the first term, $\frac{1}{1\cdot 2}$. Reporting a term as a sum is the same class of error as Devi's: a quantity that is genuinely part of the calculation, promoted to an answer it was never entitled to be.

5. (a) Both ratio-test limits. (b) Why a ratio-test limit of 1 proves nothing.

(a) For $\sum 1/n$ the ratio is $\dfrac{n}{n+1} = \dfrac{1}{1+1/n}$, whose limit is 1. For the series of reciprocal squares the ratio is $\dfrac{n^2}{(n+1)^2} = \left(\dfrac{1}{1+1/n}\right)^2$, whose limit is also

$$\boxed{L = 1}$$

(b) Open response — review by hand.
Model answer. The two series produce the same ratio-test limit and do not have the same fate: the harmonic series diverges, and the series of reciprocal squares converges. So the value $L = 1$ is compatible with both outcomes, which means it carries no information about either. A test is only evidence when different outcomes give it different values, and at $L = 1$ the ratio test has none to give. The reason is structural: the ratio test compares a series to a geometric one, and at $L = 1$ the comparison geometric series is $\sum 1^n$, which is exactly the borderline it cannot see past.
Which test settles each: the harmonic series by the p-series test with $p = 1$ (or the integral test); the series of reciprocal squares by the p-series test with $p = 2 > 1$ (or the integral test).
Look for: the explicit pairing of one convergent and one divergent example *both* giving $L = 1$; the conclusion that $L = 1$ is therefore not evidence; and a named alternative test for each. Answering only "the test is inconclusive" restates the rule without explaining it.

6. A power series is centred at $c = 2$ with radius $r = 3$. Give the two endpoints, and name the mistake behind an answer of 3.

Step 1: recall what the radius measures — a distance from the centre, not a position on the axis. The endpoints therefore sit one radius on each side of the centre: $c - r$ and $c + r$.
Step 2: compute both: $2 - 3$ and $2 + 3$:

$$\boxed{-1 \text{ and } 5}$$

Step 3: name the student's mistake. Reporting 3 as an endpoint uses the radius as though it were a coordinate — the interval was treated as though it were centred at the origin. It is the same slip as reading $|x - 2| < 3$ as $|x| < 3$; the centre shift was dropped.
Step 4: state the conclusion at the right strength. The ratio test has now given the *open* interval $-1 < x < 5$, and nothing more. Whether -1 and 5 belong is a separate question, answered in problems 7 and 8.

7. At the left endpoint the series is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. Find its exact sum, and say whether the endpoint belongs.

Step 1: choose the test the endpoint actually calls for. Substituting the endpoint has produced an alternating series, so use the alternating series test: the terms $1/n$ decrease and tend to 0, so the series converges. The convergence is conditional — the absolute series is harmonic and diverges.

Step 2: identify the value. The Maclaurin series $\ln(1+x) = \sum_{n=1}^{\infty} \dfrac{(-1)^{n-1}x^n}{n}$ evaluated at $x = 1$ gives $\ln 2$. Our series carries $(-1)^n$ rather than $(-1)^{n-1}$, so it is the negative of that: $\boxed{-\ln 2}$

Step 3: fix the sign error the problem plants. The $n = 1$ term is -1, and every partial sum stays below 0, so the sum cannot be the positive $\ln 2$. Writing the series out — $-1+\frac{1}{2}-\frac{1}{3}+\cdots$ — settles the sign in one line.

Step 4: answer the endpoint question. The series converges at $x = -1$, so the left endpoint **does belong** to the interval of convergence, and the student's open interval was wrong at that end.

8. Challenge. Write the complete interval of convergence with the correct bracket at each end, justify each end, and explain why the ratio test cannot decide an endpoint. *(Open response — review by hand.)*

Model answer.

Right endpoint. At $x = 5$ the series becomes $\sum 1/n$, the harmonic series, which diverges by the p-series test with $p = 1$. The right endpoint is excluded.

Left endpoint. At $x = -1$ the series becomes $\sum(-1)^n/n$, which converges by the alternating series test (problem 7). The left endpoint is included.

Interval: $[-1,\ 5)$ — closed on the left, open on the right. The asymmetry is the point: nothing about the ratio test predicts it, and the two ends genuinely required two different tests.

Why the ratio test cannot reach an endpoint. The ratio test concludes only when its limit is strictly less than 1 or strictly greater than 1. At either endpoint the distance from the centre equals the radius exactly, so the limit is exactly 1 — the inconclusive case from problem 5. Endpoints are, by construction, the two inputs where the ratio test is guaranteed to say nothing, which is why they must always be substituted and tested separately.

Look for: both endpoints substituted and tested by name; the correct bracket on each side; and the recognition that the endpoint limit is exactly 1 by construction. An answer that reports an open or closed interval without testing each end has committed the error the whole sheet is about, even if the bracket happens to land correctly.